Customer Purchase Behavior Analytics

A full-stack data analysis project · Python · PostgreSQL · Power BI

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Dataset	3,900 retail transactions · 18 attributes
Tools	Python (Pandas, NumPy) · PostgreSQL · Power BI

1. Executive Summary

A mid-size retail organization needed clarity on what drives customer spending, loyalty, and repeat behavior across its 3,900-transaction database. This project delivers a full-stack analytics pipeline — from raw data ingestion through SQL-based business querying to an interactive Power BI dashboard — surfacing five high-impact findings and a prioritized set of recommendations for the marketing and growth teams.

3,900	$59.76	3.75	27%	80%
Transactions Analyzed	Avg. Purchase Amount	Avg. Review Rating	Subscription Rate	Loyal Customer Base

💡 *The single largest revenue opportunity: 73% of customers (2,847 people) are not subscribed yet spend nearly identically to subscribers ($59.87 vs $59.49 avg). Converting just 10% would add ~$17,000 in tracked revenue with no change in product or pricing.*

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2. Business Question

How can customer shopping data be analyzed to uncover behavioral trends, strengthen customer engagement, and optimize marketing and product strategies?

Three sub-questions guided the analysis:

- Which customer segments and product categories generate the most revenue — and why?
- Does subscription status, discount usage, or shipping preference meaningfully affect spend?
- Where is the highest-ROI opportunity for retention, upsell, or acquisition investment?

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3. Dataset Overview

The dataset contains 3,900 retail transactions across 18 attributes covering customer demographics, product details, purchasing behavior, shipping method, discount usage, and review ratings.

Dimension	Key Fields	Notes
Customer	age, gender, location, subscription_status	Demographics & loyalty flag
Transaction	purchase_amount, season, payment_method	Core revenue fields
Product	item_purchased, category, size, color	SKU-level attributes
Behavior	discount_applied, previous_purchases, frequency_of_purchases	Loyalty & promo signals
Quality	review_rating (37 nulls)	Imputed via category-level median

💡 *Data quality was high — only 37 missing values (0.95%) in review_rating. Imputing with category-level medians preserved distributional integrity without introducing bias across product groups.*

4. Data Preparation & Transformation (Python)

Tools: Pandas · NumPy · SQLAlchemy

- Structure validation — Confirmed 3,900 rows × 18 columns; verified dtypes matched expected formats.
- Missing value handling — Located 37 nulls in review_rating. Filled using category-level medians to avoid cross-category distortion.
- Column standardization — Renamed all columns to snake_case for SQL compatibility (e.g. 'Purchase Amount (USD)' → purchase_amount).
- Redundancy removal — Confirmed discount_applied and promo_code_used were 100% correlated — dropped promo_code_used.
- Feature engineering — Created age_group (Young Adult / Adult / Middle-aged / Senior) and purchase_frequency_days field.
- Export to PostgreSQL — Loaded the cleaned DataFrame to PostgreSQL via SQLAlchemy for structured querying.

5. Data Analysis (SQL — PostgreSQL)

Fifteen business-focused queries were written across five analytical themes, using CTEs, window functions (DENSE_RANK, NTILE, running SUM OVER), and correlated subqueries.

A. Revenue & Demographic Analysis

Gender	Total Revenue	Avg. Spend	Insight
Male	$157,890	$60.25	Dominant revenue segment
Female	$75,191	$59.54	Volume gap, not a spend gap

💡 *The 2:1 revenue gap between male and female customers is driven by customer count, not spend behaviour — average purchase amounts are nearly equal ($60.32 vs $59.01). This suggests untapped potential in female customer acquisition rather than a need to change pricing or product mix.*

Revenue by Age Group

Age Group	Total Revenue	Avg. Spend
Young Adult (18–25)	$62,143	$60.45
Middle-aged (36–55)	$59,197	$60.04
Adult (26–35)	$55,978	$59.42
Senior (55+)	$55,763	$59.07

💡 *Spend per transaction is remarkably consistent across all age groups (~$59–$60). Total revenue is almost entirely a function of customer volume, not spend per head. Demographic targeting should focus on acquisition volume, not adjusting offer value by age.*

B. Product & Category Analysis

Category	Rank 1	Rank 2	Rank 3
Accessories	Jewelry (171)	Sunglasses (161)	Belt (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	—

Discount Dependency — Top 5 Products

Product	Discount Rate	Risk Level
Hat	50.0%	⚠ High
Sneakers	49.66%	⚠ High
Coat	49.07%	⚠ High
Sweater	48.17%	⚠ Medium
Pants	47.37%	⚠ Medium

💡 *Nearly 1 in 2 purchases of Hat, Sneakers, and Coat involve a discount. A price-elasticity test — reducing discount frequency by 20% on these SKUs for 60 days — would quantify true demand sensitivity before any permanent pricing change.*

C. Shipping Behavior

Shipping Type	Avg. Purchase	Observation
Express	$60.48	Higher-intent buyers
Standard	$58.46	Baseline segment

💡 *Express shipping customers spend $2.02 more per transaction on average — a self-selecting high-intent segment. Upsell and cross-sell messaging targeted at express users could yield disproportionate returns.*

D. Subscription & Loyalty Analysis

Status	Customers	Avg. Spend	Total Revenue	% of Base
Not Subscribed	2,847	$59.87	$170,436	73%
Subscribed	1,053	$59.49	$62,645	27%

💡 *Near-identical average spend confirms the subscription gap is a conversion problem, not a value problem. Non-subscribers already spend at subscriber levels — the business simply hasn't given them sufficient reason to commit. Converting 10% (~285 customers) at current spend = ~$17,000 ARR uplift.*

Customer Segmentation by Loyalty

Segment	Definition	Count	% of Base
Loyal	>10 previous purchases	3,116	79.9%
Returning	2–10 previous purchases	701	18.0%
New	≤1 previous purchase	83	2.1%

💡 *80% loyalty base is exceptional — but a 2.1% new customer share is a warning sign. Without a healthier acquisition pipeline, churn from the loyal base has no replacement.*

6. Visualization & Dashboard (Power BI)

An interactive Power BI dashboard was built to enable self-service exploration by non-technical stakeholders. All key findings are surfaced through a single-page layout with full slicer interactivity.

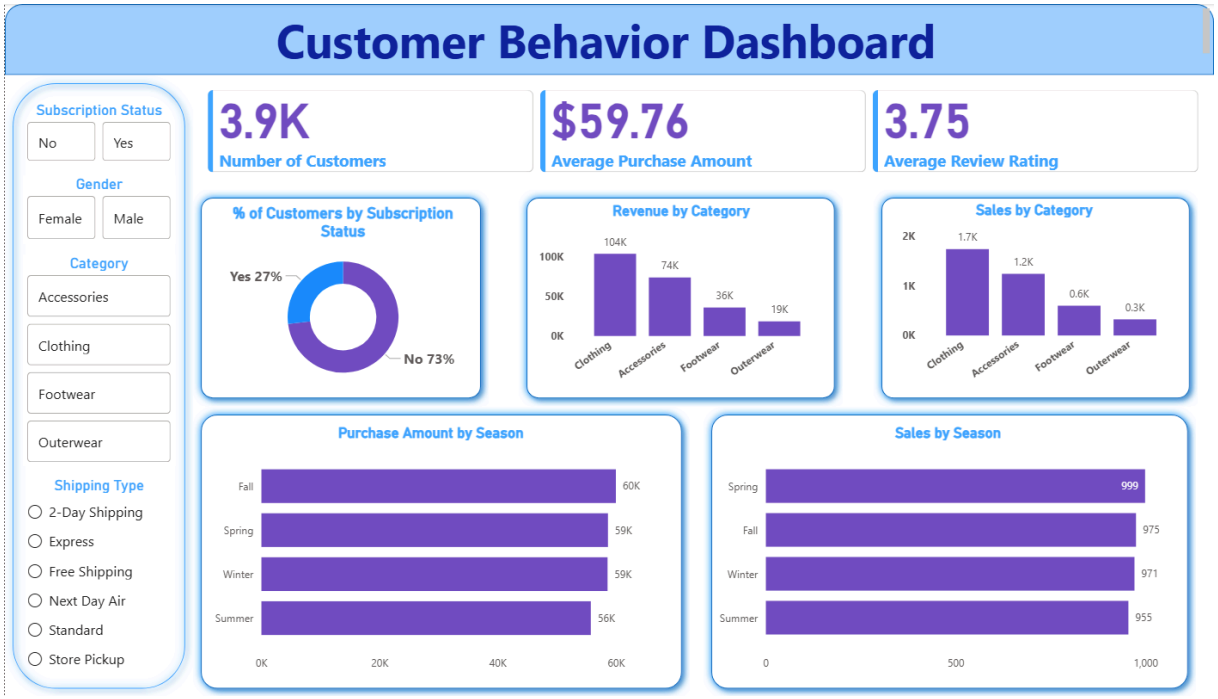

Figure 1 — Customer Behavior Dashboard (Power BI)

Visual	What It Shows	Business Use
KPI Cards	Total customers, avg purchase, avg rating	At-a-glance health metrics
Donut Chart	Subscription rate (27% Yes / 73% No)	Conversion opportunity sizing
Revenue Bar Chart	Revenue by product category	Category investment prioritisation
Season Bar Charts	Purchase amount & order volume by season	Campaign timing & inventory planning
Slicers	Subscription status, Gender, Category, Shipping type	Ad-hoc segment deep-dives

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7. Key Business Insights

Subscription gap is the #1 revenue lever

73% of customers are unsubscribed yet spend at identical per-transaction levels. The product is not the issue — the value proposition for subscribing needs strengthening.

Revenue gap is a volume problem, not a spend problem

Whether segmenting by gender, age, or subscription status, average spend per transaction is remarkably stable (~$59–$60). All major revenue differences trace back to customer count, not individual purchasing power.

Loyal customers dominate — but the funnel top is thin

80% loyalty is a strength, but 2.1% new customer share is a warning sign. Without a healthier acquisition pipeline, churn from the loyal base has no replacement.

Discount dependency is a quiet margin risk

Hat, Sneakers, and Coat are discounted in ~50% of purchases. Without price-elasticity testing, the business cannot know if these discounts drive volume or simply erode margin on sales that would have happened anyway.

Faster shipping = higher-intent buyers

Express shipping customers spend $2.02 more per order — a natural target for upsell and loyalty program enrollment.

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8. Business Recommendations

Recommendations are ranked by estimated impact and implementation ease.

Priority	Recommendation	Rationale	Expected Outcome
🔴 High	Launch a subscription conversion campaign targeting non-subscribers with >5 purchases	2,518 repeat buyers are unsubscribed; spend already at subscriber level	~$17K ARR uplift at 10% conversion
🔴 High	Introduce a tiered loyalty programme for the Loyal segment (3,116 customers)	80% of base is loyal; incremental spend increases from tier rewards are high-probability	Increased basket size + reduced churn
🟡 Medium	Run a 60-day price-elasticity test: reduce discount frequency by 20% on Hat, Sneakers, Coat	~50% discount rate on these SKUs — unclear if discounts drive volume or just erode margin	Data-driven discount policy; potential margin recovery
🟡 Medium	Increase marketing spend on Clothing and Accessories	Highest revenue and order volume in the dataset	Revenue concentration in proven performers
🟢 Lower	Design a new customer acquisition programme to grow the 2.1% new-customer share	Without new inflow, long-term revenue is exposed to loyal-base churn	Pipeline health and revenue sustainability
🟢 Lower	Create targeted Summer promotions to close the seasonal spend gap	Summer shows the lowest purchase amount ($56K) vs. Fall ($60K)	Smoother revenue distribution across seasons

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9. Tools & Technology

Layer	Tool / Library	Purpose
Data Preparation	Python 3.10 — Pandas, NumPy	Cleaning, imputation, feature engineering
Visualisation (EDA)	Matplotlib, Seaborn	In-notebook exploratory charts
Database	PostgreSQL 15	Structured querying & business analytics
BI Dashboard	Microsoft Power BI	Interactive stakeholder dashboard
Environment	Jupyter Notebook	Reproducible analysis pipeline
Version Control	Git / GitHub	Repository management & transparency